SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:25

Annexure A

Concept Mapping

Code of Conduct:

(192524467)

I Hema Priya .T *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Hemapriya

Annexure A

Concept Mapping

Question 1 (5 Marks)

Construct a multi-level concept map integrating:

Operating System Types → OS Structures → System Calls → OS Functions.

Your map must show cross-links (e.g., how OS structure influences system call implementation).

Question 2 (5 Marks)

Design a comprehensive concept map representing:

Process Management, Memory Management, File System, and I/O System.

Include interdependencies and indicate where system calls interact with each component.

Question 3 (5 Marks)

Create a comparative concept map for:

Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.

Highlight differences, similarities, advantages, and limitations using linking phrases.

Question 4 (5 Marks)

Develop a dynamic concept map showing the execution flow:

User Program → API → System Call Interface → Kernel → Hardware → Response back to User.

Include feedback loops and control flow paths.

Question 5 (5 Marks)

Construct a hierarchical + functional concept map for system calls:

Process, File, Device, and Information system calls.

Also show how they relate to OS services like scheduling, memory allocation, and device handling.

RUBRICS FOR CALCULATION

Concept Mapping

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks
Q1						
Q2						
Q3						
Q4						
Q5						
				Overall Total (100)		

Faculty In-charge	Course Coordinator	Head of Department
Signature: __________	Signature: __________	Signature: __________

QUESTION 1

PAGE 1

OPERATING SYSTEM

OS TYPES
- Batch OS
- Time-Sharing OS
- Distributed OS
- Real-Time OS

OS STRUCTURES
- Monolithic
- Layered
- Microkernel
- Hybrid

SYSTEM CALLS
- Process Mgmt
- File Mgmt
- Memory Mgmt
- Device Mgmt
- Information Mgmt

OS FUNCTIONS
- CPU Scheduling
- Memory Mgmt
- File Mgmt
- I/O & Device Mgmt
- Protection & Security

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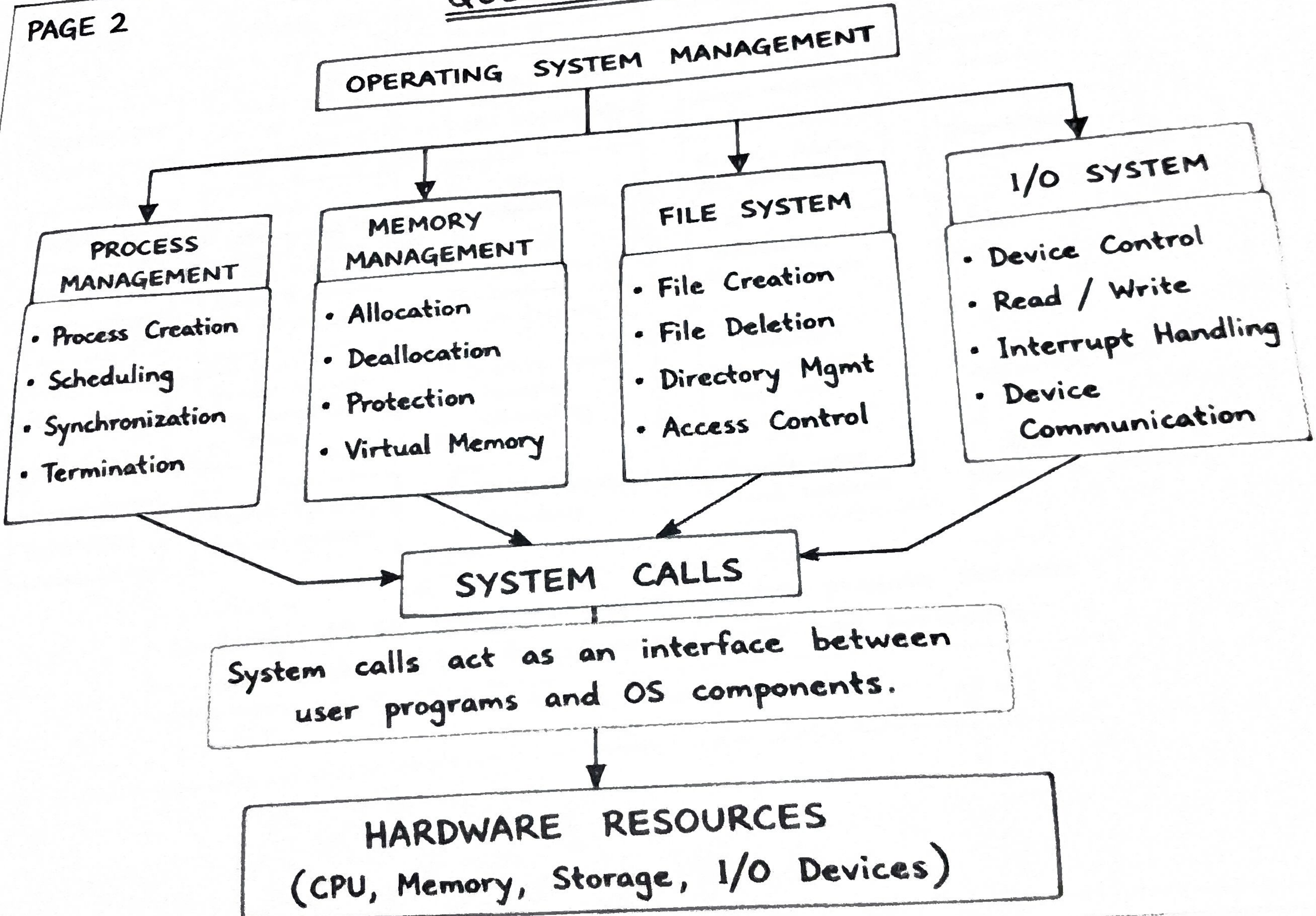
QUESTION 2



QUESTION 3

COMPARATIVE CONCEPT MAP OF OS TYPES

	BATCH OS	TIME-SHARING OS	DISTRIBUTED OS	REAL-TIME OS
DEFINITION	Executes jobs in batches without user interaction.	Allows multiple users to share CPU interactively.	Manages multiple computers and shares resources.	Provides immediate response within deadline.
KEY FEATURE	High throughput	Short response time	Resource sharing and transparency	Deterministic response
ADVANTAGES	Simple & efficient for large jobs	Better CPU utilization, user convenience	Scalability, fault tolerance	Reliability, meets deadlines
LIMITATIONS	No user interaction, long turnaround time	Security issues, complex scheduling	Complex communication, high cost	Expensive, limited application area
EXAMPLES	Payroll system, Billing system	UNIX, Linux, Windows	Google apps, Cloud systems	Robotics, Aerospace systems

Similarity: All OS types manage resources, provide services and act as an interface between user and hardware.

Date:

Date:

Date:

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QUESTION 4

EXECUTION FLOW